The mathematical relation between the numbers of bits required in the PCM code and the dynamic range is given by the expression:

$$2^n - 1 \geq DR$$

For minimum number of bits,

$$2^n - 1 = DR$$

where:

n = number of PCM bits, excluding sign bit
DR = absolute value of Dynamic Range (not dB value)

Coding Efficiency

The *coding efficiency* is defined as the numerical indication of how efficiently a PCM code is utilized. It is the ratio of the minimum number of bits required to achieve a certain dynamic range to the actual number of bits in the PCM code.

$$\text{Coding efficiency} = \frac{\text{minimum number of bits}}{\text{actual number of bits (including sign bit)}} \times 100\%$$

SQR

The Signal-to-Quantization Noise Ratio
The *signal-to-quantization noise ratio (SQR)* or just *signal-to-noise ratio (SNR)* is the ratio of the wanted signal to the unwanted noise.

$$SQR = \frac{\text{Minimum voltage}}{\text{Quantization noise voltage}}$$

This is known as the "worst-case SQR".

$$SQR = \frac{\text{Maximum voltage}}{\text{Quantization noise voltage}}$$

$$SQR(dB) = 10\log\frac{v^2/R}{(q^2/12)/R}$$

where:
R = resistance (Ω)
v = rms signal voltage (V)
q = quantization interval (V)
$\frac{v^2}{R}$ = rms signal power
$\frac{q^2/12}{R}$ = average rms quantization noise power

$$SQR(dB) = 10\log\frac{v^2}{q^2/12} = 10.8 + 20\log\frac{v}{q}$$

The PCM Coding Methods

In quantizing PAM signals into 2^n levels several methods are used. These are:
- Level-at-a-time coding
- Digit-at-a-time coding
- Word-at-a-time coding

Level-at-a-time Coding
In *level-at-a-time coding*, the PAM signal is compared to a ramp waveform while a binary counter is being advanced to a uniform rate. When the ramp waveform equals or exceeds the PAM sample, the counter contains the PCM code.

Digit-at-a-time Coding
In *digit-at-a-time coding*, each digit of the PCM code is determined sequentially. It is similar to a balance where known reference weights are used to determine an unknown weight.

Word-at-a-time Coding
These coders are flash encoders and are more complicated than earlier coders. They are useful for high-speed applications.

Companding Methods

Companding in general as term implies, is the process of *compressing* and *expanding*. This process helps the signal and minimizes the effect of noise.

Analog Companding

Analog companding was popular in older PCM systems. It was implemented using specially designed diodes. The analog compressor is inserted between bandpass filter and sample-and-hold circuit in the transmitter.

The two popular analog companding methods in use are:
- μ-Law companding (used in US and Japan)
- A-Law companding (used in Europe established by ITU-T)

μ-Law Companding

In μ-law companding, the compression characteristic is given by

$$V_{out} = \frac{V_{max} \times \ln(1+\mu V_{in}/V_{max})}{\ln(1+\mu)}$$

where:
V_{max} = maximum uncompressed analog input amplitude
V_{in} = amplitude of the input signal at a particular instant of time

A-Law Companding

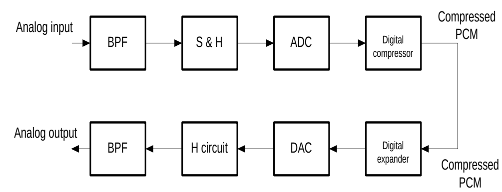
In A-law companding, the compression characteristic is given by

$$V_{out} = V_{max}\frac{A V_{in}/V_{max}}{1+\ln A} \quad ; 0 \leq \frac{V_{in}}{V_{max}} \leq \frac{1}{A}$$

$$V_{out} = V_{max}\frac{1+\ln(A V_{in}/V_{max})}{1+\ln A} \quad ; \frac{1}{A} \leq \frac{V_{in}}{V_{max}} \leq 1$$

Digital Companding

In *digital companding*, the compression is done after the input sample is converted to a linear PCM code (transmitter side) and the expansion is done before PCM encoding (receiver side).



Differential Pulse Code Modulation

DPCM solves the problem of redundancy (resulting to more bits transmitted) in a typical PCM-encoded speech waveform. It is designed specifically to take advantage of the sample-to-sample redundancies.
In DPCM, the difference in the amplitude of two successive samples is transmitted rather than the actual sample thus fewer bits are required to be transmitted since the range of sample differences is less than the range of individual samples.

Delta Modulation (DM)
Delta modulation utilizes a single-bit PCM code. It uses single bit PCM code which simply indicates + or - in magnitude.

Problems of Delta Modulator

Slope overload
When the slope of the analog signal is greater than the delta modulator can keep, slope overload occurs. This problem can be remedied by increasing the clock frequency and the magnitude of the minimum step size.

Granular noise
Granular noise in delta modulation is similar to quantization noise in PCM system. This occurs when the reconstructed signal has variations that were not present in the original signal due to the relatively constant amplitude of original analog input. This problem is minimized by decreasing the step size.